

Vectorisation of the IKOB algorithm

A. Dimensions:

n_z = number of zones, $n_i = 4$ (income classes), $n_s = 5$ (urbanity levels), n_g
= number of groups, n_v = number of transport modes

Typographic conventions:

Boldface uppercase denotes matrices (e.g. $\mathbf{M}$), boldface lowercase denotes vectors (e.g. $\mathbf{v}$), and regular italic denotes scalars (e.g. α).

Operations:

$\odot$ denotes the Hadamard (element-wise) product, $\oslash$ denotes element-wise division, $\mathbf{1}_n$ is the all-ones column vector of length n , and $\text{diag}(\mathbf{v})$ constructs a diagonal matrix from vector $\mathbf{v}$. The notation $\mathbf{M}[\mathbf{s}, :]$ means row-indexing of matrix $\mathbf{M}$ by an integer index vector $\mathbf{s}$ (i.e., row z of the result is row s_z of $\mathbf{M}$). Broadcasting of a column vector $\mathbf{v} \in \mathbb{R}^{n_z}$ across all columns of a matrix $\mathbf{M} \in \mathbb{R}^{n_z \times n_i}$ is written as $\text{diag}(\mathbf{v}) \mathbf{M}$. The element-wise maximum of a set of matrices of equal shape is written $\max_v \{\cdot\}$.

B. Distribution of Groups over Neighborhoods — Vectorised

Given data structures

$$\mathbf{AZ}^{\text{tab}} \in \mathbb{R}^{n_s \times n_i}, \mathbf{GA}^{\text{tab}} \in \mathbb{R}^{n_s \times n_i}, \mathbf{GR}^{\text{tab}} \in \mathbb{R}^{n_s \times n_i}$$

These are Tables 1, 2, and 3 respectively (percentages divided by 100). Furthermore:

$$\mathbf{P} \in \mathbb{R}^{n_z \times n_i}: \text{income-class proportions per zone}$$

$$\mathbf{s} \in \{1, \dots, 5\}^{n_z}: \text{urbanity grade vector}$$

$$\mathbf{aa} \in \mathbb{R}^{n_z}: \text{actual cars per household per zone}$$

Step 1: Car ownership — vectorised

Row-selection by urbanity. For each zone z , select the row from each table corresponding to urbanity s_z :

$$\widetilde{\mathbf{AZ}} = \mathbf{AZ}^{\text{tab}}[\mathbf{s}, :] \in \mathbb{R}^{n_z \times n_i}$$

$$\widetilde{\mathbf{GA}} = \mathbf{GA}^{\text{tab}}[\mathbf{s}, :] \in \mathbb{R}^{n_z \times n_i}$$

$$\widetilde{\mathbf{GR}} = \mathbf{GR}^{\text{tab}}[\mathbf{s},:] \in \mathbb{R}^{n_z \times n_i}$$

Theoretical zone-level values (income-class-specific):

$$\mathbf{AZ}_{iz}^{\text{theor}} = \widetilde{\mathbf{AZ}} \odot \mathbf{P} \in \mathbb{R}^{n_z \times n_i}$$

Theoretical zone totals (summed over income classes):

$$\mathbf{az}_{\text{theor}} = (\widetilde{\mathbf{AZ}} \odot \mathbf{P}) \mathbf{1}_{n_i} \in \mathbb{R}^{n_z}$$

$$\mathbf{ga}_{\text{theor}} = (\widetilde{\mathbf{GA}} \odot \mathbf{P}) \mathbf{1}_{n_i} \in \mathbb{R}^{n_z}$$

$$\mathbf{gr}_{\text{theor}} = (\widetilde{\mathbf{GR}} \odot \mathbf{P}) \mathbf{1}_{n_i} \in \mathbb{R}^{n_z}$$

Correction for actual car ownership. Define the correction factors element-wise:

$$\mathbf{f}_{\text{auto}} = \mathbf{aa} \oslash \mathbf{az}_{\text{theor}} \in \mathbb{R}^{n_z}$$

$$\mathbf{f}_{\text{no}} = (\mathbf{1}_{n_z} - \mathbf{aa}) \oslash (\mathbf{1}_{n_z} - \mathbf{az}_{\text{theor}}) \in \mathbb{R}^{n_z}$$

Corrected per-zone per-income-class values (broadcasting the correction vector across income classes):

$$\mathbf{AZ}_{iz} = \text{diag}(\mathbf{f}_{\text{auto}}) (\widetilde{\mathbf{AZ}} \odot \mathbf{P}) \in \mathbb{R}^{n_z \times n_i}$$

$$\mathbf{GA}_{iz} = \text{diag}(\mathbf{f}_{\text{no}}) (\widetilde{\mathbf{GA}} \odot \mathbf{P}) \in \mathbb{R}^{n_z \times n_i}$$

$$\mathbf{GR}_{iz} = \text{diag}(\mathbf{f}_{\text{no}}) (\widetilde{\mathbf{GR}} \odot \mathbf{P}) \in \mathbb{R}^{n_z \times n_i}$$

Note: the correction factor $\mathbf{f}_{\text{auto}}$ is only applied where $\mathbf{aa} < \mathbf{az}_{\text{theor}}$ (element-wise). This conditional can be expressed using an indicator vector $\mathbf{m} = \mathbb{1}[\mathbf{aa} < \mathbf{az}_{\text{theor}}]$ and blending:

$$\mathbf{f}_{\text{auto}} = \mathbf{m} \odot (\mathbf{aa} \oslash \mathbf{az}_{\text{theor}}) + (\mathbf{1}_{n_z} - \mathbf{m}) \odot \mathbf{1}_{n_z}$$

and correspondingly:

$$\mathbf{f}_{\text{no}} = \mathbf{m} \odot ((\mathbf{1}_{n_z} - \mathbf{aa}) \oslash (\mathbf{1}_{n_z} - \mathbf{az}_{\text{theor}})) + (\mathbf{1}_{n_z} - \mathbf{m}) \odot \mathbf{1}_{n_z}$$

Step 2: Free car and free public transport — vectorised

Given $\mathbf{grA} \in \mathbb{R}^{n_i}$ (Table 4, company-car percentages per income class):

$$\mathbf{GrA}_{iz} = \mathbf{AZ}_{iz} \text{diag}(\mathbf{grA}) \in \mathbb{R}^{n_z \times n_i}$$

$$\mathbf{WA}_{iz} = \mathbf{AZ}_{iz} - \mathbf{GrA}_{iz} \in \mathbb{R}^{n_z \times n_i}$$

Given $\mathbf{grOV}^{\text{tab}} \in \mathbb{R}^{n_s}$ (Table 5, free-OV percentage per urbanity):

$$\mathbf{grOV} = \mathbf{grOV}^{\text{tab}}[\mathbf{s}] \in \mathbb{R}^{n_z}$$

Step 3: Transport mode preferences — vectorised

Given $\mathbf{VK}^{\text{tab}} \in \mathbb{R}^{n_s \times n_v}$ (Table 6) and $\mathbf{VK}^{\text{nocar}} \in \mathbb{R}^{n_s \times (n_v - 1)}$ (Table 7):

$$\mathbf{VK} = \mathbf{VK}^{\text{tab}}[\mathbf{s}, :] \in \mathbb{R}^{n_z \times n_v}$$

$$\mathbf{VK}_{\text{nocar}} = \mathbf{VK}^{\text{nocar}}[\mathbf{s}, :] \in \mathbb{R}^{n_z \times (n_v - 1)}$$

C. Distribution of Jobs over Target Groups — Vectorised

Given $\mathbf{S}_{\text{LISA}} \in \mathbb{R}^{n_z \times n_{\text{SBI}}}$ (number of jobs per zone per SBI category), $\mathbf{E} \in \mathbb{R}^{n_{\text{SBI}} \times n_{\text{opl}}}$ (education-level distribution per SBI, the table on page 8), and $\mathbf{Q} \in \mathbb{R}^{n_{\text{opl}} \times n_i}$ (education-to-income mapping, the table on page 8):

$$\mathbf{A} = \mathbf{S}_{\text{LISA}} \mathbf{E} \mathbf{Q} \in \mathbb{R}^{n_z \times n_i}$$

where entry A_{bi} gives the number of jobs in zone b belonging to income class i . This is a straightforward chain of two matrix multiplications that maps SBI sectors $\rightarrow$ education levels $\rightarrow$ income classes.

D1. Experienced Travel Time — Vectorised

Data structures

Travel time and distance data are stored as origin–destination matrices:

$$\mathbf{T}_v \in \mathbb{R}^{n_z \times n_z}: \text{in-vehicle travel time by mode } v$$

$$\mathbf{D}_v \in \mathbb{R}^{n_z \times n_z}: \text{in-vehicle distance by mode } v$$

Zone-level attributes:

$$\mathbf{pza} \in \mathbb{R}^{n_z}: \text{parking search time at arrival (destination attribute)}$$

$$\mathbf{pzv} \in \mathbb{R}^{n_z}: \text{parking search time at departure (origin attribute)}$$

For OV additionally:

$$\mathbf{VT} \in \mathbb{R}^{n_z}: \text{pre-transport time per origin}$$

$\mathbf{NT} \in \mathbb{R}^{n_z}$: post-transport time per destination

$\mathbf{T}_o \in \mathbb{R}^{n_z}$: transfer time at boarding station

$\mathbf{T}_u \in \mathbb{R}^{n_z}$: transfer time at alighting station

$\mathbf{W} \in \mathbb{R}^{n_z \times n_z}$: waiting/transfer time matrix

Pure travel time matrices

Auto:

$$\mathbf{R}_{\text{auto}} = \mathbf{p} \mathbf{z} \mathbf{v} \mathbf{1}_{n_z}^{\top} + \mathbf{1}_{n_z} \mathbf{p} \mathbf{z} \mathbf{a}^{\top} + \mathbf{T}_{\text{auto}} \in \mathbb{R}^{n_z \times n_z}$$

This constructs the full origin–destination matrix by broadcasting the origin-specific departure search time across columns, the destination-specific arrival search time across rows, and adding the in-vehicle time.

OV (with bicycle as access/egress):

$$\mathbf{R}_{\text{ov}} = \mathbf{V} \mathbf{T} \mathbf{1}_{n_z}^{\top} + \mathbf{T}_o \mathbf{1}_{n_z}^{\top} + \mathbf{T}_{\text{ov}} + \mathbf{W} + \mathbf{1}_{n_z} \mathbf{N} \mathbf{T}^{\top} + \mathbf{1}_{n_z} \mathbf{T}_u^{\top} \in \mathbb{R}^{n_z \times n_z}$$

(Electric) bicycle:

$$\mathbf{R}_{\text{fiets}} = \mathbf{T}_{\text{fiets}} \in \mathbb{R}^{n_z \times n_z}$$

Total cost matrices (per group)

For a group g with variable cost $k_{\text{var},g}$, toll rate $k_{\text{hef},g}$, parking cost $k_{\text{park},g}$, and cordon charge $k_{\text{cor},g}$:

Auto:

$$\mathbf{K}_{g,\text{auto}} = (k_{\text{var},g} + k_{\text{hef},g}) \mathbf{D}_{\text{auto}} + k_{\text{park},g} \mathbf{J} + k_{\text{cor},g} \mathbf{J}$$

where $\mathbf{J} = \mathbf{1}_{n_z} \mathbf{1}_{n_z}^{\top}$ is the all-ones matrix (or more precisely, a matrix encoding whether each OD pair crosses a cordon; for simplicity this can be an element-wise indicator matrix).

OV:

$$\mathbf{K}_{g,\text{ov}} = k_{\text{var},g,\text{ov}} \mathbf{D}_{\text{ov}} + k_{\text{opstap}} \mathbf{J} + k_{\text{park},g,\text{fiets}} \mathbf{J}$$

Fiets: $\mathbf{K}_{g,\text{fiets}} = \mathbf{0}$ (zero matrix).

Experienced travel time matrix (per group, per mode)

Let τ_i be the time value of money for income class $i(g)$:

$$\mathbf{ER}_{g,v} = \mathbf{R}_v + \tau_{i(g)} \mathbf{K}_{g,v} \in \mathbb{R}^{n_z \times n_z}$$

This is a pure element-wise scaled addition of two origin–destination matrices.

D2. From Experienced Travel Time to Weights — Vectorised

The logistic travel-time-decay function is applied element-wise to the entire OD matrix. Define the element-wise sigmoid:

$$\sigma(\mathbf{M}) = (\mathbf{J} + \exp(\mathbf{M}))^{\odot -1}$$

where $\exp(\cdot)$ is the element-wise exponential and $(\cdot)^{\odot -1}$ is element-wise reciprocal.

For group g with preference p , transport mode v , and motif m , and with parameters α_{pvm} , ω_{pvm} , and w_{pvm} from Tables 9–11:

$$\mathbf{G}_{g,v,m} = w_{pvm} \sigma(\alpha_{pvm}(-\omega_{pvm} \mathbf{J} + \mathbf{ER}_{g,v})) \in \mathbb{R}^{n_z \times n_z}$$

Expanded element-wise:

$$[\mathbf{G}_{g,v,m}]_{hb} = \frac{w_{pvm}}{1 + \exp(\alpha_{pvm}(-\omega_{pvm} + [\mathbf{ER}_{g,v}]_{hb}))}$$

This is a single vectorised pass: one scalar multiplication, one matrix addition, one element-wise exponential, one element-wise addition, one element-wise reciprocal, and one scalar multiplication.

D3. From Single-Mode Weights to Combined Weights — Vectorised

For a given combination of transport modes $\mathcal{V} \subseteq \{1, \dots, n_v\}$, the combined weight matrix is the element-wise maximum over the modes in the combination:

$$\mathbf{GC}_{g,m}^{(\mathcal{V})} = \max_{v \in \mathcal{V}} \mathbf{G}_{g,v,m} \in \mathbb{R}^{n_z \times n_z}$$

If the single-mode weight matrices are stacked into a three-dimensional tensor $\mathcal{G}_{g,m} \in \mathbb{R}^{n_z \times n_z \times |\mathcal{V}|}$, then:

$$\mathbf{GC}_{g,m}^{(\mathcal{V})} = \max_{\text{axis}=3} \mathcal{G}_{g,m}$$

This is a reduction (max) along the transport-mode axis of the stacked tensor.

D4. From Combined Weights to Reachable Jobs — Vectorised

Per group

Let $\mathbf{a}_i \in \mathbb{R}^{n_z}$ be the column vector of jobs for income class i (column i of matrix $\mathbf{A}$ from Section C). Let $\mathbf{v}_g \in \mathbb{R}^{n_z}$ be the population vector of group g (number of people in group g per zone). Let $i(g)$ be the income class to which group g belongs.

The reachable-jobs vector for group g is:

$$\mathbf{b}_g = \mathbf{v}_g \odot (\mathbf{G}\mathbf{C}_{g,m} \mathbf{a}_{i(g)}) \in \mathbb{R}^{n_z}$$

Here $\mathbf{G}\mathbf{C}_{g,m} \mathbf{a}_{i(g)}$ is a matrix–vector product ($n_z \times n_z$ times $n_z \times 1$), yielding the weighted job count per origin zone. The Hadamard product with $\mathbf{v}_g$ scales by group population.

Per income class

Sum over all groups belonging to income class i :

$$\mathbf{b}_i = \sum_{g \in \mathcal{G}_i} \mathbf{b}_g \in \mathbb{R}^{n_z}$$

If the group population vectors for all groups in $\mathcal{G}_i$ are stacked as columns of $\mathbf{V}_i \in \mathbb{R}^{n_z \times |\mathcal{G}_i|}$, and the per-group reach vectors as columns of $\mathbf{R}_i \in \mathbb{R}^{n_z \times |\mathcal{G}_i|}$ (where each column is $\mathbf{G}\mathbf{C}_{g,m} \mathbf{a}_i$), then:

$$\mathbf{b}_i = (\mathbf{V}_i \odot \mathbf{R}_i) \mathbf{1}_{|\mathcal{G}_i|}$$

Regional aggregation

Let $\mathbf{m}_r \in \{0,1\}^{n_z}$ be the indicator vector for zones in region r , and $\mathbf{I}_i \in \mathbb{R}^{n_z}$ the inhabitants vector for income class i :

$$B_{i,r} = \frac{(\mathbf{b}_i \odot \mathbf{I}_i)^{\square\top} \mathbf{m}_r}{\mathbf{I}_i^{\square\top} \mathbf{m}_r}$$

This is a population-weighted average of the per-zone reach, restricted to the zones in region r .

D5. From Combined Weights to Reachable Inhabitants — Vectorised

Per group

Here the matrix is traversed in the transpose direction. Let $\mathbf{i}_g \in \mathbb{R}^{n_z}$ be the number of inhabitants in group g per zone:

$$\mathbf{b}_g^{\text{inh}} = \mathbf{G}\mathbf{C}_{g,m}^{\square\top} \mathbf{i}_g \in \mathbb{R}^{n_z}$$

Entry b of this vector gives the weighted number of inhabitants (of group g) that can reach destination zone b .

Per income class

$$\mathbf{b}_i^{\text{inh}} = \sum_{g \in \mathcal{G}_i} \mathbf{b}_g^{\text{inh}} \in \mathbb{R}^{n_z}$$

Equivalently, if we collect $\mathbf{i}_g$ for all $g \in \mathcal{G}_i$ as columns of $\mathbf{I}_{\mathcal{G}_i} \in \mathbb{R}^{n_z \times |\mathcal{G}_i|}$, and the corresponding combined-weight matrices as pages of a tensor, then the income-class reach can be written as a sum of matrix–vector products. If all groups within an income class share the same combined weight matrix $\mathbf{GC}_{i,m}$ (which happens when groups within an income class differ only in non-spatial attributes already absorbed into the weight matrix), this simplifies to:

$$\mathbf{b}_i^{\text{inh}} = \mathbf{GC}_{i,m}^{\square\top} \left(\sum_{g \in \mathcal{G}_i} \mathbf{i}_g \right) = \mathbf{GC}_{i,m}^{\square\top} \mathbf{I}_i$$

Regional aggregation (weighted by jobs)

$$B_{i,r}^{\text{inh}} = \frac{(\mathbf{b}_i^{\text{inh}} \odot \mathbf{a}_i)^{\square\top} \mathbf{m}_r}{\mathbf{a}_i^{\square\top} \mathbf{m}_r}$$

D6. Competitive Position of Neighborhoods — Vectorised

The competitive position measures, for each origin zone, how favourably placed it is relative to other zones competing for the same jobs. The key idea is to weight each destination's jobs by the inverse of the "potential" (number of reachable inhabitants) at that destination.

Per group

Define the attraction-ratio vector:

$$\mathbf{r}_g = \mathbf{a}_{i(g)} \oslash \mathbf{b}_g^{\text{inh}} \in \mathbb{R}^{n_z}$$

where $\mathbf{b}_g^{\text{inh}}$ is from D5 (reach in inhabitants per destination for group g). Then:

$$\mathbf{c}_g = \mathbf{GC}_{g,m} \mathbf{r}_g \in \mathbb{R}^{n_z}$$

This is a single matrix–vector product. Entry h of $\mathbf{c}_g$ is the competitive position of origin zone h for group g .

Per income class

$$\mathbf{c}_i = \frac{\sum_{g \in \mathcal{G}_i} \mathbf{c}_g \odot \mathbf{v}_g}{\sum_{g \in \mathcal{G}_i} \mathbf{v}_g} \in \mathbb{R}^{n_z}$$

In matrix form, with $\mathbf{C}_i \in \mathbb{R}^{n_z \times |\mathcal{G}_i|}$ (columns are $\mathbf{c}_g$) and $\mathbf{V}_i \in \mathbb{R}^{n_z \times |\mathcal{G}_i|}$ (columns are $\mathbf{v}_g$):

$$\mathbf{c}_i = (\mathbf{C}_i \odot \mathbf{V}_i) \mathbf{1}_{|\mathcal{G}_i|} \oslash \mathbf{V}_i \mathbf{1}_{|\mathcal{G}_i|}$$

Regional aggregation

$$C_{i,r} = \frac{(\mathbf{c}_i \odot \mathbf{I}_i)^{\top} \mathbf{m}_r}{\mathbf{I}_i^{\top} \mathbf{m}_r}$$

D7. Competitive Position of Job Zones — Vectorised

This is the "mirror" of D6, measuring how favourably a destination zone competes for workers.

Per group

Define the supply-pressure vector:

$$\mathbf{q}_g = \mathbf{i}_g \oslash (\mathbf{G}\mathbf{C}_{g,m} \mathbf{a}_{i(g)}) \in \mathbb{R}^{n_z}$$

The denominator is the reachable-jobs vector from D4 (before scaling by $\mathbf{v}_g$). Then:

$$\mathbf{c}_g^{\text{dest}} = \mathbf{G}\mathbf{C}_{g,m}^{\top} \mathbf{q}_g \in \mathbb{R}^{n_z}$$

Entry b of $\mathbf{c}_g^{\text{dest}}$ gives the competitive position of destination zone b for group g .

Per income class

$$\mathbf{c}_i^{\text{dest}} = \sum_{g \in \mathcal{G}_i} \mathbf{c}_g^{\text{dest}} \in \mathbb{R}^{n_z}$$

Regional aggregation (weighted by jobs)

$$C_{i,r}^{\text{dest}} = \frac{(\mathbf{c}_i^{\text{dest}} \odot \mathbf{a}_i)^{\top} \mathbf{m}_r}{\mathbf{a}_i^{\top} \mathbf{m}_r}$$

Summary of Computational Kernels

The entire IKOB pipeline, once vectorised, reduces to a small set of repeated computational kernels across groups and transport-mode combinations:

The **outer-product additions** in D1 construct the OD travel-time and cost matrices from zone-attribute vectors. The **element-wise sigmoid** in D2 applies the travel-time-decay function to an entire OD matrix in one pass. The **element-wise maximum** in D3 compresses several per-mode OD weight matrices into a single combined weight matrix. The **matrix-vector products** in D4–D7 are the core operation: multiplying the $n_z \times n_z$ combined-weight matrix by a destination-attribute vector (D4, D6) or transposing it and multiplying by an origin-attribute vector (D5, D7). The **Hadamard products and reductions** aggregate over groups within an income class and over zones within a region.

Because the weight matrix $\mathbf{GC}_{g,m}$ is shared across D4–D7, it need only be computed once per group/motif. The dominant cost is $\mathcal{O}(n_z^2)$ per group per mode for the matrix–vector products, and $\mathcal{O}(n_z^2)$ for each sigmoid evaluation — both of which are highly parallelisable on modern hardware (GPU, SIMD).